

IPL Franchise Valuation & Player ROI

Project Documentation & Architecture

1. Project Overview

This project is an interactive, "Moneyball"-style sports analytics dashboard built in Power BI. It evaluates the financial efficiency of Indian Premier League (IPL) franchises by comparing a player's auction acquisition cost against their actual on-field performance (runs scored and wickets taken). The dashboard serves as a comprehensive tool for analyzing squad building strategies and tracking capital allocation efficiency.

2. Business Problem & Objective

During the IPL auction, franchises spend massive budgets (totaling upwards of ₹1,192 Crores across the league) to build competitive squads. However, high auction prices do not always guarantee high performance, frequently leading to inefficient capital allocation.

The primary objective of this dashboard is to move beyond traditional cricket statistics and accurately measure **Financial Return on Investment (ROI)**. It enables team management, scouts, and data analysts to:

- **Identify Draft Steals:** Pinpoint undervalued players who provide high output at a low acquisition cost.
- **Expose Inefficiencies:** Highlight expensive underperformers who consume large portions of the salary cap with minimal on-field contribution.
- **Audit Franchise Strategy:** Evaluate aggregate spending efficiency at the team level to inform future auction bidding strategies.

3. Analytical Methodology

To accurately measure financial efficiency, the data model normalizes player valuation by isolating specific roles. Evaluating a pure bowler based on runs scored creates skewed, inaccurate data. Therefore, the architecture relies on context-specific DAX (Data Analysis Expressions) measures engineered directly into the Power BI semantic model.

By dividing monetary spend by distinct performance units, the model creates an equitable baseline to compare players across different price tiers and franchise rosters.

Custom DAX Engineering

- **Cost_Per_Run_₹** : Evaluates Batters and All-rounders by dividing their total auction price by total runs scored. Identifies the true cost of batting output.
- **Cost_Per_Wicket_₹** : Evaluates Bowlers and All-rounders by dividing their total auction price by total wickets taken. Normalizes bowling efficiency independent of batting expectations.

4. Dashboard Features & Functionality

The dashboard provides a top-down executive view that seamlessly drills down into individual player analytics via interconnected visuals.

- **Interactive Tile Navigation:** Users can filter the entire dashboard architecture by selecting specific Franchise tiles (e.g., Chennai Super Kings, Mumbai Indians) or Player Roles (Batter, Bowler, All-rounder, Wicketkeeper) using clean, app-style slicers.
- **Dynamic KPI Scorecard:** A centralized metric ribbon displays real-time, responsive calculations for Total Spend (₹ Cr), Total Runs, and Total Wickets based strictly on the selected franchise or player role.
- **Dual-Pillar ROI Scatter Plots:**
 - *Batting ROI (Price vs. Runs):* Maps batters across a quadrant to instantly visualize high-performing bargains (bottom-right) versus costly investments (top-left).
 - *Bowling ROI (Price vs. Wickets):* Mirrors the batting visual to map bowling efficiency, guaranteeing pure bowlers are correctly assessed on their primary skill.
- **Performance Matrix:** A comprehensive data table providing granular, player-by-player breakdowns of aggregate output and exact monetary cost per unit of performance.

5. Technical Stack

- **Platform:** Microsoft Power BI Desktop
- **Data Engineering & ETL:** Power Query (Data type validation, handling nulls, formatting)
- **Calculations:** DAX (Data Analysis Expressions) for custom ROI metrics
- **UI/UX Design:** Dark-theme executive layout, dynamic conditional filtering, classic KPI scorecards